

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Inference Quiz

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01 AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Analyze and select appropriate agent strategies for different types of environments and problem settings (BTL 4)*

Assessment Tool Weightage: 20%

Total Marks: 50

Time : 45 Mins

Each question carries equal marks.

No.	Question	Option A	Option B	Option C	Option D	Correct Answer	Time (Sec)
1	A system rule: "If A then B" results in what when A is true but B is not observed?	Alarm OFF	System error	No definite conclusion	Alarm ON	C – No definite conclusion	45
2	Rule: $A \wedge B \rightarrow C$ evaluates to what when A is TRUE and B is FALSE?	TRUE only if A false	TRUE	Cannot be determined	FALSE	D – FALSE	45
3	Expression: $\neg(P \rightarrow Q)$ is equivalent to which formula?	$P \vee Q$	$\neg P \wedge \neg Q$	$\neg P \vee \neg Q$	$P \wedge \neg Q$	D – $P \wedge \neg Q$	45
4	Clauses: $(\neg A \vee B)$ and C. What can be deduced?	Nothing	Only B and C	Only B	Only C	D – Only C	45

5	Unify: $P(x, f(y))$ and $P(a, f(y))$. Which substitution unifies?	$\{x=a\}$	$\{x=b, y=a\}$	$\{x=a, y=b\}$	No unification	$A - \{x=a\}$	45
6	Unify: $P(x, x)$ with $P(a, b)$. What is the result?	Loop	Failure	Partial	Success	$B - \text{Failure}$	50
7	Convert $\forall x (P(x) \rightarrow \neg P(x) \vee Q(x))$ to CNF components. Which is correct?	$\neg Q(x) \wedge P(x)$	$P(x) \vee Q(x)$	$P(x) \wedge \neg Q(x)$	$P(x) \wedge Q(x)$	$B - P(x) \vee Q(x)$	45
8	Convert: $\exists x \forall y P(a, y)$. Which propositional form matches?	$\forall y P(x, y)$	$\exists x P(a, x)$	None	$\forall y P(a, y)$	$D - \forall y P(a, y)$	45
9	Given $P \rightarrow Q$, what additional information is required to conclude Q ?	C	Q and R	P	None	$C - P$	50
10	Forward chaining next inferred fact from A?	All A,B,C	C	B	A	$C - B$	40
11	Backward chaining: to prove A, which are subgoals?	None	B	A	A and B	$B - B$	40
12	Predicate: Loves. Which category does "Constant" belong to?	Variable	Predicate	Function	Constant	$D - \text{Constant}$	45
13	Ground term: What is a ground instance of $W(x)$?	$P(x)$	John	$f(x)$	C	$D - C$	45
14	Convert $\neg(P \rightarrow Q)$ into CNF. Which is correct?	$\neg P \wedge Q$	$\neg P \vee Q$	$P \vee Q$	$P \wedge \neg Q$	$D - P \wedge \neg Q$	50

15	KB: Rain $\rightarrow$ Wet, $\neg$ Wet. What follows?	None	C	Both	Slippery	A – None	50
16	Inference engine	Reasoning	Output	Input	Storage	A –	55

	component responsible for choosing rules is what?					Reasoning	
17	Graph reasoning: Which nodes are expanded next from B?	C	B,C	D	B,C,D	B – B,C	55
18	Conflict: Bird $\rightarrow$ Flies conflicts with another rule. What is chosen?	Random	Priority rules	Delete KB	Ignore	B – Priority rules	50
19	Resolution eliminates which units during simplification?	Variables	Literals	Clauses	Functions	B – Literals	50
20	Predicate: Parent. Which is a propositional variable?	Constant	Function	Predicate	Variable	D – Variable	55
21	Expression: $P \wedge (P \wedge Q) \vee R$ equals which form?	$(P \wedge Q) \vee R$	$(P \wedge Q) \wedge R$	Same	$(P \vee Q) \wedge (P \vee R)$	A – $(P \wedge Q) \vee R$	45
22	DNF of $P \wedge (Q \vee (P \wedge Q) \vee (P \wedge R))$ simplifies to what?	None	$P \vee Q \vee R$	$P \wedge (Q \vee R)$	$(P \vee Q) \wedge (P \vee R)$	C – $P \wedge (Q \vee R)$	45
23	Inference rule: If P is true, which is correct?	P false	None	Q true	B	B – None	50
24	Semantic Network: “Dog is cold-blooded” implies what?	Dog cannot bark	Dog is warm-blooded	Dog is not a mammal	None of the above	D – None of the above	45

25	Frame: Vehicle has how many wheels?	6	4	2	Unknown	D – Unknown	45
26	Unify: $P(x,y)$ and $P(a,John)$ result is?	$\{x=John\}$	$\{x=a, y=John\}$	$\{y=John\}$	No substitution	$B - \{x=a, y=John\}$	50
27	Resolution: $\neg P \vee P \vee Q$	Q	$\neg Q$	None	B	$A - Q$	50

	resolves with what?						
28	Logic Gate: $A=True$ and $B=False$ gives output?	UNKNOWN	ERROR	FALSE	TRUE	$C - FALSE$	45
29	Which of the following is equivalent to $P \rightarrow Q$?	$P \vee Q$	$P \wedge Q$	$\neg P \vee Q$	$\neg Q \wedge P$	$C - \neg P \vee Q$	45
30	Graph: $A \rightarrow B \rightarrow C$. Which nodes are reachable from A?	Only B	B and C	B,C,D	Only D	$B - B$ and C	45
31	Unify: $P(x,x)$ and $P(a,b)$ outcome is?	Success	Infinite loop	Partial	Failure	$D - Failure$	45
32	Skolemization: $\exists x P(x)$ becomes which form?	B	$\neg P(x)$	$\forall x P(x)$	$P(a)$	$D - P(a)$	45
33	Medical rule: $Fever \rightarrow Flu$. If there is no fever then?	No conclusion	Flu diagnosed	Fever false	Rule ignored	$A - No$ conclusion	50

34	Smart irrigation: If soil is wet, then pump is OFF. If soil is dry, pump is ON. If soil is wet, what happens?	Error	Pump ON	No action	Pump OFF	D – Pump OFF	55
35	Backward chaining starts from which element?	Goal	Facts	Data	Rules	A – Goal	50
36	Forward chaining begins with which element?	Ignored	Stops	Facts	Starts from goal	C – Facts	40
37	KB conflict: Birds fly, Penguins don't.	Random	Conflict resolution	Statistical	Deducti on	B – Conflict resolution	40

	Which strategy should be used?						
38	Chatbot: Hungry user. Which action should be performed first?	Suggest food	Delete rule	Ignore	Store data	A – Suggest food	50
39	Unify: Eat(x,food) with Eat(John,apple) gives?	{food=John}	{x=apple}	{x=John, food=apple}	No match	C – {x=John, food=apple}	50
40	Autonomous car: Green light → Accelerate. When the light is red?	Ignore	Stop	Turn left	Acceler ate	B – Stop	50
41	What is the result of unifying F(a) and F(b)?	{a=a}	{b=a}	{a=b}	No unificat ion	D – No unificatio n	45

42	In a directed graph, if A points to B and C, which nodes can be reached from A?	Only C	None	B and C	Only B	C – B and C	50
43	Which is a valid conclusion from “If it rains, then the ground is wet”?	It is not raining	The ground is wet	None of the above	It is raining	C – None of the above	40
44	Unify: Likes(x,food) with Likes(Emily,pizza) results in?	No match	{x=Emily, food=pizza}	{x=pizza}	{food=Emily}	B – {x=Emily, food=pizza}	40
45	Which can be concluded from “If the light is on, then the room is bright”?	The light is off	None of the above	The room is bright	It is dark	B – None of the above	45
46	Forward chaining begins with which element?	Data	Facts	Rules	Goal	B – Facts	50
47	$A \vee (B \wedge C)$, when A=FALSE, B=TRUE, C=FALSE, gives what?	TRUE	Depends on C	Cannot be determined	FALSE	D – FALSE	45
48	In backward chaining, if you want to prove B, which must be established?	A	C	A and C	None	A and C	40
49	Given “If A then B,” what can be inferred if B is observed?	Cannot be determined	A is false	Rule is valid	A is true	A – Cannot be determined	45

50	Unify: Likes(x,ice_cream) and Likes(Mary,chocolate) results in?	{x=Mary, ice_cream= chocolate}	No match	{ice_cream=Mary}	{x=chocolate}	B – No match	40
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Code of Conduct:

I K Sathwik Kumar (192425051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Sathwik Kumar

Q.No	Correct Answer
1	C – No definite conclusion
2	D – FALSE
3	D – $P \wedge \neg Q$
4	D – Only C
5	A – $\{x=a\}$
6	B – Failure
7	B – $P(x) \vee Q(x)$
8	D – $\forall y P(a,y)$
9	C – P
10	C – B
11	B – B
12	D – Constant
13	D – C (as printed in the PDF)
14	D – $P \wedge \neg Q$
15	A – None
16	A – Reasoning
17	B – B,C
18	B – Priority rules
19	B – Literals
20	D – Variable
21	A – $(P \wedge Q) \vee R$
22	C – $P \wedge (Q \vee R)$
23	B – None
24	D – None of the above

25	<i>D – Unknown</i>
26	<i>B – {x=a, y=John}</i>
27	<i>A – Q</i>
28	<i>C – FALSE</i>
29	<i>C – $\neg P \vee Q$</i>
30	<i>B – B and C</i>
31	<i>D – Failure</i>
32	<i>D – P(a)</i>
33	<i>A – No conclusion</i>
34	<i>D – Pump OFF</i>
35	<i>A – Goal</i>
36	<i>C – Facts</i>
37	<i>B – Conflict resolution</i>
38	<i>A – Suggest food</i>
39	<i>C – {x=John, food=apple}</i>
40	<i>B – Stop</i>
41	<i>D – No unification</i>
42	<i>C – B and C</i>
43	<i>C – None of the above</i>
44	<i>B – {x=Emily, food=pizza}</i>
45	<i>B – None of the above</i>
46	<i>B – Facts</i>
47	<i>D – FALSE</i>
48	<i>A and C</i>
49	<i>A – Cannot be determined</i>
50	<i>B – No match</i>

RUBRICS FOR CALCULATION:

Each correct answer is of 1 mark.